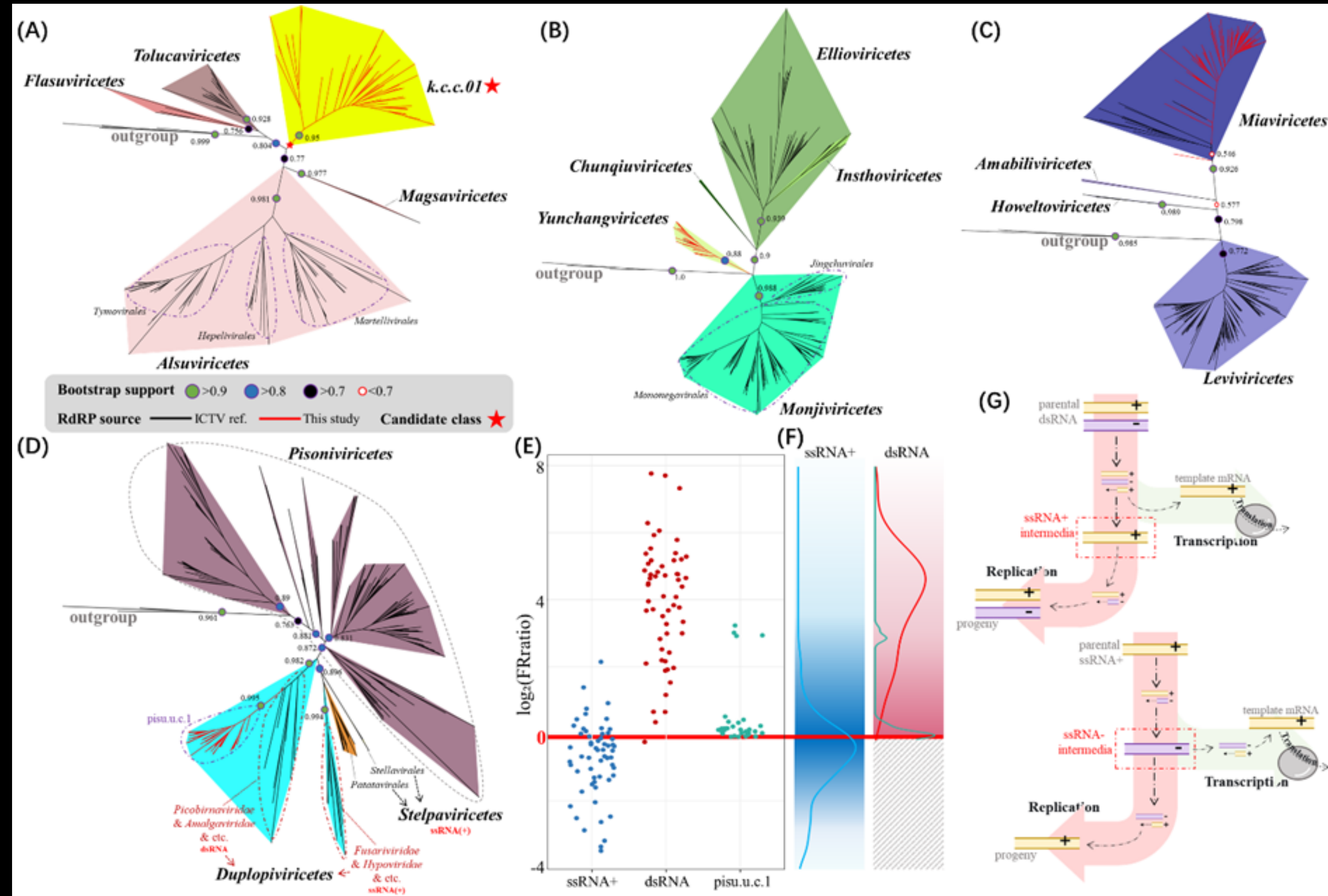


Phylogenetic Reconstruction and Strand-Specific Signatures of Established RNA Virus Phyla

RdRP-based phylogenies resolve class-level structure across major RNA virus phyla, while strand-specific logFRratio analyses distinguish ssRNA+ and dsRNA viruses consistent with their replication biology.



Directional Selection Shapes Amino Acid Exchange Landscapes Across RNA Virus Phyla

Site-specific analyses reveal heterogeneous amino acid exchange patterns consistent with diverse selective pressures, with globally distributed directional selection signals exemplified in *Lenarviricota*.

